Programming Fundamentals (CS1002)
Mid 1 Solution

Note = operator has been used for assignment while the == operator has been used here to check equality of two operands.

Sr.	Pseudocode	Output

a.	`Declare value, p, q`
	`p = 25`
	`q = 7`
	`While q > 0`
	`  value = (p MOD q)`
	`  p = q`
	`  q = value`
	`End While`
	`Print p`
		1

b.	`Declare x, y, z`
	`x = 1, y = 2, z = 3`
	`x = y + z`
	`z = x - y`
	`z = z + x`
	`print x, y, z`
	`z = y + z`
	`y = y + z`
	`Print x, y, z`
		5 2 8
		5 12 8

c.	`Declare sum = 0, i = 1`
	`While i <= 5`
	`  If i mod 2 == 0 then`
	`  sum = sum + i`
	`  Else`
	`  sum = sum + (i * 2)`
	`  End If`
	`End While`
	`Print sum`
		Infinite Loop
		
		OR
		
		24

d.	`Declare x, y`
	`x = 12, y = 7`
	`If x > 10 then`
	`  If y < 5 then`
	`  result = x + y`
	`  Else if y < 10 then`
	`  result = x - y`
	`  Else`
	`  result = x * y`
	`  End If`
	`Else if x == 10 then`
	`  result = y * 2`
	`Else`
	`  result = 0`
	`End If`
	`Print result`
		5

e.	`Declare n, i`
	`n = 6`
	`While i < n do`
	`  If i mod 2 == 0 then`
	`  Print "i = ", i, " Square = ", i * i`
	`  Else`
	`  Print "i = ", i, " Cube = ", i * i * i`
	`  End If`
		Ideally:
		i = 0 Square = 0
		i = 1 Cube = 1
		i = 2 Square = 4
		i = 3 Cube = 27
		i = 4 Square = 16
		i = 5 Cube = 125
		i = 0 Square = 0

	`End While`	i = 0 Square = 0 i = 0 Square = 0 i = 0 Square = 0 …. Infinite loop

Q2	CLO 2	15 Marks

A university determines student admission based on the entrance test marks, HSSC (FSc or equivalent) percentage, and SSC (Matric or equivalent) percentage.

The minimum eligibility requirements are:

- At least **60% in SSC (or equivalent)**.
- At least **50% in HSSC (or equivalent)**.

If a student fails to meet these requirements, the system should **inform them exactly what they are lacking**:

- If SSC < 60%, output *"Not Eligible: SSC below 60% requirement."*
- If HSSC < 50%, output *"Not Eligible: HSSC below 50% requirement."*
- If both are below the required threshold, output **both reasons**.

An aggregate score in admissions is basically a combined percentage that reflects a student's performance across different components (like Admission Test, HSSC, SSC), based on assigned weightages.

If the candidate meets the eligibility requirements, the aggregated score is computed using weightages of admission test, HSSC, and SSC:

- Weightage of Admission Test marks 50%
- Weightage of HSSC/equivalent 40%
- Weightage of SSC/equivalent 10%

The admission outcome is then decided as follows:

- Aggregate ≥ 65% → *"Seat Granted"*.
- 60% ≤ Aggregate < 65% → *"Wait for possible 2nd merit list"*.
- Aggregate < 60% → *"Did not meet merit"*.

Additionally, any candidate scoring more than 90% in the aggregate is awarded a scholarship and a *"100% Scholarship Awarded"* message is displayed on screen. If the candidate does not qualify for the scholarship, no message should be displayed about it.

All three inputs (SSC%, HSSC%, Test Marks) are to be taken at the start.

Task:

Draw a flowchart that takes SSC percentage, HSSC percentage, and Admission Test marks as input, determines eligibility with specific reasons if ineligible, and if eligible, computes the aggregate and outputs the admission decision (and scholarship if applicable).

	Sample Input	**Sample Output**
1	Admission Test Marks: 40 HSSC Percentage: 48 SSC Percentage: 70	Not Eligible: HSSC below 50% requirement.
2	Admission Test Marks: 85 HSSC Percentage: 92 SSC Percentage: 88	Candidate is Eligible. Seat Granted 100% Scholarship
3	Admission Test Marks: 60 HSSC Percentage: 51 SSC Percentage: 60	Candidate is Eligible. Did not meet merit

SOLUTION:

Correct Inputs and Outputs	02 Marks
Aggregate Calculation	03 Marks
Correct Output	10 Marks

Q3	CLO 2	15 Marks

The user will enter **N numbers**. Write a pseudocode to determine the **count** of the **longest consecutive series of numbers entered** in which each number is greater than the previous number by exactly 1. The program should then display this count as the output.
First, ask the user how many numbers they want to enter (let's call this N). Then, take input of N numbers from the user, and try to find the longest consecutive series of numbers.
The following sample runs are given for your convenience, so the statement above is easy for you to understand.

	Sample Inputs	**Sample Output**	**Explanation**
1	N = 10 Numbers: 1, 2, 11, 5, 0, 23, 24, 25, 29, 31	**3**	*You take 10 inputs because N is 10.* *3: Because The sequence 23, 24, 25 consists of three consecutive numbers, which is the longest.*
2	N = 8 Numbers: 4, 5, 2, 4, 7, 8, 9, 10	**4**	*You take 8 inputs because N is 8.* *Because The sequence 7, 8, 9, 10 consists of four consecutive numbers, which is the longest, even though it occurs later in the sequence.*

SOLUTION:

Correct input	01 Mark
Loop runs N times	04 Marks
Correct streak calculation	10 Marks

```
START
  DECLARE curr_len, N, max_len, count
  PRINT "How many numbers (N)?"
  READ N
  PRINT "Enter the numbers one by one:"
  READ prev                  // first number
  curr_len ← 1
  max_len  ← 1
  count    ← 1               // we already read one number

  WHILE count < N DO
      READ x
      count ← count + 1

      IF x = prev + 1 THEN
          curr_len ← curr_len + 1
      ELSE
          curr_len ← 1
      ENDIF

      IF curr_len > max_len THEN
          max_len ← curr_len
      ENDIF

      prev ← x
  ENDWHILE

  PRINT max_len
END
```